Salting the Earth: Shrimp Farming and Agricultural Externalities in Bangladesh

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Abstract

Climate change is raising groundwater salinity across coastal regions globally, threatening smallholder livelihoods. In southern Bangladesh, a common adaptation is to convert freshwater rice paddies into saltwater shrimp ponds. However, this adaptation comes with an externality: saltwater from shrimp ponds may seep into neighboring fields, damaging rice. This paper considers whether such private adaptation generates negative externalities on neighboring rice farmers and whether those externalities accelerate further adoption. Combining a novel household survey with a satellite panel of 73,300 plots over nine years, we document that a neighbor's conversion reduces vegetation health on adjacent rice plots and raises own conversion probability by 18.0 percentage points. Interpreted through a model of land use based on comparative advantage and a boundary externality, these results imply an adaptation cascade: 51 percent of ponds exist only because a neighbor converted, and a tax on conversion would raise aggregate farm profit by 10 percent. Private decisions that promote climate resilience for some can erode it for others.

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1 Introduction

It is well established that human activity is altering the global climate (IPCC, 2021). A large body of evidence shows this is harming livelihoods, especially in the world's poorest countries (Dell et al., 2012; Burke et al., 2015; Carleton et al., 2022). The extent to which these changes affect human populations depends, in large part, on people's ability to adapt to new conditions (IPCC, 2022).

One setting where adaptation may be slow is agriculture. The cost of new technology, barriers to learning, and heterogeneous returns are just a few well-documented obstacles to take-up.[1] Since agricultural workers are especially exposed to the climate through both physical exposure and the relationship between heat and yields, slow adoption may be especially costly in this sector. In this paper, we argue that one countervailing force that may accelerate adoption is a non-user production externality. If adopting a new technology harms non-users who can avoid that damage by adopting it themselves, take-up may be faster than in the absence of the externality.

We study this dynamic in the context of southwestern Bangladesh, where farmers can respond to climate-change-induced groundwater salinity by switching from rice cultivation (the old technology) to inland shrimp farming (the new technology) in saltwater ponds. During the dry season, farmers can flood their plots with salty water from canals or slow-flowing rivers, which protects them from the damage that saltwater intrusion would otherwise have done to their rice. However, this behavior imposes an externality on their neighbors who choose not to adopt saltwater shrimp farming: rice farmers who choose not to adopt face even higher salinity not only because of climate change, but also because of their neighbors' adoption decisions.

First, using a novel survey that combines a household questionnaire (N = 739) with ground-truth salinity readings (N = 1,478), we document three stylized facts that support the paper's argument that the salinity externality exists and affects patterns of adoption. *Fact 1:* We begin by showing that farmers believe shrimp farming generates a production externality. In a hypothetical plot exercise, farmers expect that, in the year following their neighbors' conversion from rice to shrimp, their own plots will experience both higher salinity and lower yields equivalent to a 48 percent decrease in output if they do not adopt shrimp farming as well. This yield loss estimate is nearly two orders of magnitude larger than the average annual decrease in rice yields due to climate change-induced salinity alone, which suggests that damages from shrimp farming are first order for rice farmers in this context

[1] For surveys of this literature, see Magruder (2018); Suri and Udry (2022); Adamopoulos et al. (2026)

(Dasgupta et al., 2015). *Fact 2:* Farmers' beliefs about salinity are supported by the cross-sectional evidence. Rice plots bordering shrimp ponds have salinity readings 20.6 percent (0.6 deciSiemens per meter) higher than those of plots that do not border shrimp ponds. We take readings in May, immediately before the onset of the annual rainy period, and show that shrimp-adjacent plots have higher readings than their non-shrimp-adjacent counterparts. *Fact 3:* Finally, these saltier rice plots that border shrimp ponds produce lower rice yields (171 kg/acre less) than plots that do not border shrimp farms. While this is not causal evidence of an externality, it is notable that even in equilibrium, when farmers are responding to their neighbors, they are unable to offset these losses and equalize yields across adjacent plots, even if they are able to choose salt-tolerant rice varieties. Taken together, these facts suggest that there is scope for farmers to make land-use decisions with this production externality in mind.

Second, to study whether farmers' beliefs about this externality affect patterns of adoption, we construct a new plot-level panel of land use covering 73,300 plots in coastal Bangladesh from 2017 to 2025. We apply Delineate Anything, a frontier deep-learning model for field-boundary delineation, to high-resolution PlanetScope imagery, allowing us to recover individual plot polygons and the adjacency network through which spillovers may propagate. We then classify each plot during *Rabi* (the dry season, which lasts from November to February) as agriculture or aquaculture using a seasonal median water index from Sentinel-2 imagery, with data-driven thresholds that require no manual calibration.[2] To our knowledge, this is the first plot-level, seasonal record of shrimp aquaculture adoption at this scale.

Third, we use the resulting panel to estimate the effects of a neighbor's conversion to aquaculture in an event-study framework, using the estimator of Sun and Abraham (2021) to account for staggered conversion timing and heterogeneous treatment effects across cohorts. A key identification challenge is spatially correlated shocks: both plots may be responding to the same rising underlying salinity. To address the risk of correlated shocks, we tile the study region with hexagons that average 20 plots per hexagon and control for hexagon-by-time fixed effects. Using this approach, we show that the conversion of a plot to aquaculture induces an 18.0 percentage point increase in the likelihood that a bordering plot adopts aquaculture as well. This effect persists for at least five years until the end of the sample. Using the same empirical strategy, we further show that among plots that remain in agriculture, a satellite-

[2]We apply the Modified Normalized Difference Water Index (MNDWI) of Xu (2006). This index better separates water from built-up land and is more reliable in detecting shallow water compared to NDWI.

derived measure of crop health falls by 1 percent, or 3 percent of the baseline mean, in the year following a neighbor's conversion, before returning to pre-conversion levels in subsequent years. Because this estimate is obtained only from plots that themselves never convert to aquaculture, it likely understates the average penalty: if the most severely affected plots are the most likely to exit agriculture, the remaining sample is positively selected. Together, these results are consistent with an environmental externality that not only damages neighboring agriculture but also reshapes land-use decisions, accelerating the spread of aquaculture.

Fourth, to interpret these effects, we develop a model of land-use choice under a production externality. We build on the Roy (1951) model of selection on comparative advantage in which profits from either shrimp or rice depend on observable plot characteristics and a Fréchet-distributed productivity for every crop à la Costinot et al. (2016); Sotelo (2020) and Farrokhi and Pellegrina (2023). However, in these models, a parcel's returns do not depend on neighbors' crop choices. We add an environmental externality that acts only at the boundary between two adjoining plots to mirror salinity spillovers through groundwater. This lowers the threshold at which a plot switches to aquaculture, creating the potential for a spatial cascade. We model the cascade as a static repeated problem because the empirical analysis gives no evidence of anticipation and land-use decisions are year-over-year reversible; thus, there are no option value gains to waiting to adopt. Last, because beliefs about damages are two orders of magnitude larger than what climate change alone would imply, we do not explicitly model a global salinity trend. The event study estimate of the share of plots that adopt shrimp farming due to neighbors' behavior identifies the size of the net externality favoring shrimp farming under the assumption that profits observed in the household survey pin down the dispersion of plots' productivity draws. We estimate the model from the cross-section of rice and shrimp farms in the household survey and the conversion event study from the satellite panel. The dispersion of realized output identifies the shape of the distribution of comparative advantage, while plots' land-use choices identify how size and distance to the river shift relative suitability, with the baseline rate of aquaculture anchored to the satellite data. Equipped with the estimated model, we can compare the take-up of shrimp among plots where we set the externality to zero. On a stylized landscape, we find that the externality generates a substantial cascade: 51 percent of the plots that take up aquaculture do so only because of the externality from their neighbors.

The model and event studies together only allow us to speak to whether or not there is a cascade. To assess whether the observed extent of shrimp farming represents

over- or under-adaptation, we decompose the net spillover estimated from land-use choices into damage to neighboring rice and benefits to neighboring ponds, using farmers' elicited beliefs about the rice yield loss a neighbor's conversion would cause, and compare the resulting allocation with the planner's allocation. The result of this exercise is that the recently announced subsidy regime for shrimp in Bangladesh (8% for exports) is in the wrong direction—our model implies that the level of shrimp farming adoption in southern Bangladesh is too high and that a tax on the conversion of agricultural land to aquaculture would increase welfare by 10 percent.

Contributions: A large literature studies the drivers of technology adoption in agriculture. Economists have examined this question since at least Griliches (1957), who showed that the diffusion of hybrid corn tracked its profitability, and subsequent work has identified many reasons adoption may be slow even when new technologies are profitable on average. Farmers must learn about a technology's returns, either through their own experience or from their neighbors (Foster and Rosenzweig, 1995; Conley and Udry, 2010); returns are heterogeneous, so low adoption may partly reflect low returns for some farmers (Suri, 2011); and farmers may fail to notice relevant dimensions of a technology or learn from misspecified models (Hanna et al., 2014). In developing countries, a further set of frictions can slow adoption: missing credit and insurance markets (Karlan et al., 2014; Mobarak and Rosenzweig, 2013; Casaburi and Willis, 2018), exposure to downside risk (Emerick et al., 2016), factor market failures (Jones et al., 2022), behavioral biases (Duflo et al., 2011), and limited access to information (Beaman et al., 2021; Aker and Jack, 2025). We study a setting in which an externality imposed on non-adopters instead accelerates adoption. We show that plots only respond to other plots with which they share a direct border, farmers also report that they discuss farming with their direct neighbors no more often than with other households in the village, and rice greenness declines on neighboring plots that never convert. Together, these patterns suggest the results are not driven by learning alone. In this setting, the new technology (i.e., shrimp farming) acts as a force that speeds adoption, even though these other frictions would, on net, lead us to expect slow uptake.

Second, we contribute to the literature on adaptation to climate change in low- and middle-income countries.[3] This literature has largely focused on whether and how households adapt and on the constraints that limit adaptation. We highlight a different dimension of heterogeneity in the consequences of adaptation: adaptive behavior by some households can make others worse off. In our setting, conversion to

[3]See Kala et al. (2026) and Carleton et al. (2024) for reviews.

aquaculture accelerates further adoption, while also imposing costs on those who do not convert. In this respect, our paper relates to work showing that adaptation can generate costs for others, such as groundwater depletion from competitive pumping (Ryan and Sudarshan, 2022; Blakeslee et al., 2020) and conflict arising from pastoralist migration in response to drought (McGuirk and Nunn, 2025). Relative to this work, we observe the externality directly at the level of individual plots and their neighbors, allowing us to trace how damage to one plot affects its owner's subsequent decisions and how adoption spreads across a landscape. We show that in responding to climate shocks, people can exacerbate the very shocks for others that they themselves sought to avoid.

Third, within the climate adaptation literature, we contribute to a growing body of work on responses to salinity. Rising salinity in coastal Bangladesh has been linked to migration (Chen and Mueller, 2018), economic precarity (Chen et al., 2022), early-life health (Guimbeau et al., 2024), and lower returns to high-yielding rice varieties (Dasgupta et al., 2015), while recent work studies the adoption of salt-tolerant rice (Patel, 2025). Closest to our work, Haque (2025) shows that fragmented land markets and religious homogeneity can impede the adoption of shrimp aquaculture, but that land may be reallocated toward shrimp through the rental market. A related study documents higher salinity and lower paddy profits on plots closer to shrimp ponds (Morshed et al., 2020). We complement this work with a plot-level seasonal panel tracking land use across 73,300 plots over nine years, which allows us to study adoption dynamically rather than at a single point in time. By observing plot boundaries and the adjacency network between plots, we show that the externality from conversion can generate a cascade in which each adoption raises the likelihood of further conversions nearby.

The remainder of the paper proceeds as follows. Section 2 introduces the setting. Section 3 discusses the data. Section 4 introduces the model. Section 5 presents the stylized facts. Section 6 presents the empirical strategy and event-study results. Section 7 discusses estimation of the model and the counterfactuals. Finally, Section 8 concludes.

2 Context/Setting

This section provides background information on the key features of southwestern Bangladesh where this study takes place. It discusses the salinity conditions of the setting and the extent to which saltwater shrimp farming can be accuratly described

as climate adaptation.

2.1 Climate Change in Bangladesh

This project takes place in the Khulna and Satkhira districts of southwestern Bangladesh. Appendix Figure B1 shows the locations of unions (the smallest administrative unit of Bangladesh) in which we collect survey data; unions used in the satellite analysis are shown in pink. Both districts are bounded by mangrove forests and the saltwater Bay of Bengal to the south, and inland Bangladesh to the north. These districts sit within the Ganges-Brahmaputra-Meghna delta and are low-lying and densely populated. A large share of households work in agriculture, and rice dominates (Bangladesh Bureau of Statistics, 2022). In addition to other climate-induced shocks such as heatwaves, droughts, and flooding that affect the entire country, people living in these districts face disparate additional threats such as storm surges and saltwater intrusion World Bank Group (2024). Among these threats, salinity is notable in that it builds gradually and directly reduces agricultural productivity.

Two different processes may govern the evolution of soil salinity, operating on two different time scales. The first is a slow regional change driven by climate and upstream water use. Sea-level rise and storm surges push saline water further inland, while reduced freshwater flows in the dry season allow tidal salinity to travel further up coastal rivers and permeate groundwater (Dasgupta et al., 2015). According to the Bangladesh Ministry of Environment, Forest, and Climate Change, the area under critical saltwater intrusion of 5–10 parts per thousand is expected to increase by 68% by 2050 (Ministry of Environment, Forest and Climate Change, Government of the People's Republic of Bangladesh, 2018). This is a long-run, ongoing process: the Soil Resource Development Institute in the Bangladesh Ministry of Agriculture has found that from 1973 to 2009, salt-affected areas increased by 26.7% (Soil Resource Development Institute, 2010). The second process is both faster and more localized. When a plot is converted to aquaculture, saline water is brought onto the plot and can spread to neighboring fields within a season, as we show in Section 6. This process is shaped, in party by distance to the coast and by infrastructure: the system of polders and embankments built from the 1960s was designed to keep saline water out of agricultural land, but shrimp farmers have been documented breaching embankments to bring saline water inland (Brammer, 1990). The regional trend thus gradually changes which plots are close to the margin between rice and aquaculture, while the local externality determines how conversion spreads among neighbors within much shorter periods.

Policies that may limit salinity exist, but often require coordinated management that has been difficult to achieve in coastal Bangladesh. A key tool to address groundwater intrusion is the enforcement of limits on over-pumping informed by regular water balance assessments (Stein et al., 2023). Other approaches that facilitate recharge of groundwater include shifting demand to other water sources, reducing water consumption, switching to salt-tolerant crops, or otherwise changing production processes. Because groundwater is a common-pool resource, regulation requires central authority; however, de facto water governance suffers from weak coordination among agencies and ineffective local water management institutions (Esha et al., 2026). Even if central policy were agreed upon, smallholder water users may also resist limits, which drives up the cost of monitoring and enforcement.

2.2 Shrimp Farming as Climate Adaptation

Shrimp farming in Bangladesh involves flooding ponds with saltwater, generally drawn from the sea or saline rivers, at high tide and retaining it at low tide (Morshed et al., 2020). This takes place largely on paddy fields or salt pans on tidal flats, ringed by low earthen levees that are often built inside government flood-control polders. At the end of a growing season, salt ponds may be pumped out to allow for agriculture or to introduce new saline water for a new season of shrimp. In this setting, mixed-use plots are common; representative sampling suggests approximately two-thirds of shrimp plots are used intermittently for rice cultivation (Mondal et al., 2013). The most commonly farmed species is the black tiger shrimp (Penaeus monodon, locally *bagda*), which is stocked at low densities, with little or no external feed and reliance on tidal water exchange, so the shrimp largely subsist on the natural plankton and organisms that develop in the pond. The critical input is the seed stock, which is purchased from a network of hatcheries across coastal areas.

Figure 1 shows benchmark ranges for salt tolerance of the crops considered in this paper. For non-tolerant rice, we use the standard threshold of 3 dS/m (soil saturation extract), above which yield declines by about 12 percent per additional dS/m (Maas and Hoffman, 1977). These non-salt-tolerant varieties remain the most common varieties in the dry season in the area of our study (Islam et al., 2024). For salt-tolerant rice, we use 8 dS/m, the upper end of the whole-season tolerance of BRRI dhan67 (Paul et al., 2024). Newer rice varieties such as BRRI dhan97 and BRRI dhan99 are available at higher salinity tolerances, but produce lower average yields and are still rarely adopted, making up less than 1 percent of current use (Debsharma et al., 2024).

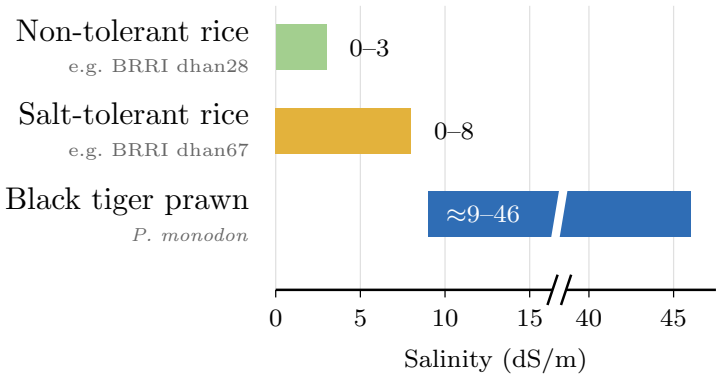


Figure 1. Salinity Range for Rice and Shrimp

Black tiger prawns (*Penaeus monodon*) tolerate water salinity of roughly 5–35 ppt, and farming in Bangladesh typically takes place at 5–20 ppt (Rahi et al., 2021). In controlled trials with Bangladeshi stock, growth and survival were highest at 20 ppt and lower but viable at 5 ppt (Rahi et al., 2021). We therefore take 5–30 ppt as the range over which shrimp culture is feasible. Because shrimp requirements refer to water salinity, we convert them to electrical conductivity at 25 °C using the Practical Salinity Scale (UNESCO, 1981), which places this range at approximately 9–46 dS/m. The comparison is therefore between soil salinity for rice and water salinity for shrimp and should be read as indicative. On these benchmarks, shrimp culture becomes feasible at a salinity just above the whole-season tolerance of the most widely grown salt-tolerant rice variety.

3 Data

This section describes the data used in the paper.

3.1 Household Survey

The household agricultural survey was conducted between May and June of 2025 and covered 739 households across fourteen unions from the Khulna and Satkhira districts of Bangladesh. To select the sample, enumerators first identified marginal plot triples within villages where they could observe an active saltwater shrimp pond adjacent to two agricultural plots. This construction (i.e., a central agricultural plot with a saline shrimp pond on one side and another agricultural plot on the other side) allowed us to collect salinity readings which could provide a gradient at both an agriculture–agriculture boundary and an agriculture–aquaculture boundary. After

selecting the plots, they identified the plot operators to interview. After obtaining consent and completing the modules with the operators, they returned to the plots to take salinity readings.

3.2 Salinity Readings

Salinity readings were collected using a portable salinity meter with a range of 0–10 mS/cm. The meter has a metal probe that is inserted directly into moist soil, where it passes a small electrical current between its tips and measures how easily that current flows. Because dissolved salts carry electricity, saltier soil conducts the current more readily, and the meter converts this into a salinity reading shown on its screen, which is then entered into the survey by an enumerator. The primary advantage of this type of meter is that it provides a nearly instantaneous reading and requires minimal training to operate. However, it is important to note that the meters measure overall electrical conductivity, which will depend on soil moisture, texture, and compactness. They have been shown to be less accurate than lab tests and can fail if soil contact is poor. To improve quality, we calibrated all meters using a standardized saline solution before the survey was conducted and took each reading twice. We average the two readings before using them in the analyses presented in this paper.

May was chosen for the timing of the survey because it was immediately before the beginning of the monsoon season and thus gave us an opportunity to observe salinity at or near its peak, following its accumulation throughout the dry winter period (Rahman et al., 2025). The salinity data presented here are therefore best interpreted as showing a gradient between saline and non-saline plots at or near its greatest extent.

3.3 Satellite Data

Our analysis of satellite imagery combines two data sources to first delineate plots, and then categorize those plots as either agricultural land or aquaculture.

Field boundaries are delineated using preprocessed high-resolution Planet basemaps, which are single seamless images that aggregate cloud-free pixels from repeated satellite passes over a common area. Because this imagery is essentially a mosaic of different time periods, it can be used to recognize permanent and semi-permanent features of the landscape, such as the boundaries between plots, but cannot be used to categorize the use of the plot during any season. At Bangladesh's latitude, the pixels captured by the sensors on the satellite will correspond to 4.4 meters. We process this imagery by applying the Delineate Anything model, a machine learning approach

that performs instance segmentation (i.e., identifies individual plots, rather than the lines between plots). The algorithm is pre-trained on a sample of 73 million plots across the world and is discussed further in Appendix C. Because the mosaics are already cloud composites, they are an ideal sample on which to extract plot geometry. We retain the resulting field polygons as the fixed spatial units of the panel. Here, we abstract away from the division of one plot into multiple plots or the combination of multiple plots into one larger plot during the span of the satellite portion of our analysis (i.e., 2017–2025). Panels 1 and 2 of Figure 2 show this first step. Panel 1 displays the basemap mosaic. Panel 2 shows the delineated plots in this setting.

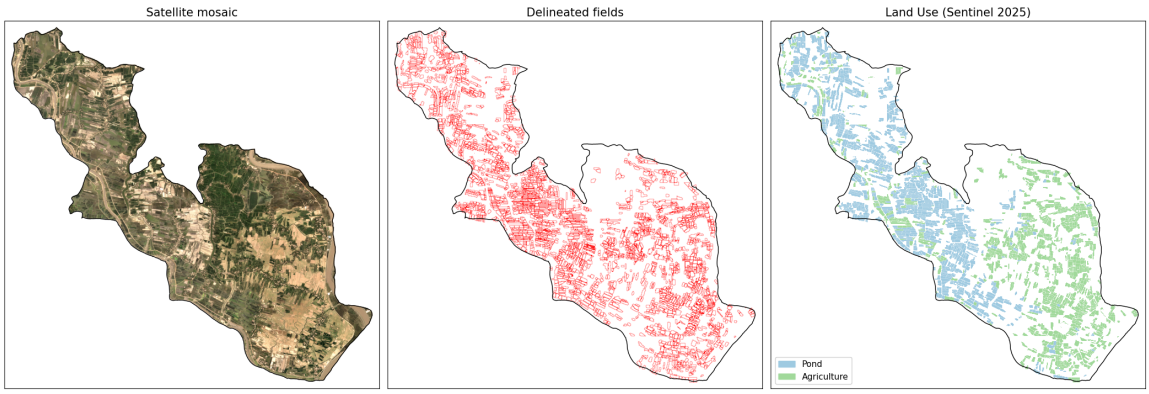


Figure 2. Illustrative Plot Classification - Magurkhali Union

To measure land use over time, we use Sentinel-2 satellite imagery, a lower-resolution but higher-frequency dataset from a European satellite program with a spatial resolution of 10 to 20 meters per pixel depending on the spectral band. For every plot from 2017 to 2025, we compute two indices from the Sentinel-2 bands. The first is the Modified Normalized Difference Water Index (MNDWI), a standard formula combining a green and a shortwave-infrared band that produces high values over standing water and low values over vegetation and soil, which we use to detect ponds.[4] The second is the Normalized Difference Vegetation Index (NDVI), likewise a standard greenness measure that is high over healthy vegetation and low over bare or waterlogged ground, which we use as a proxy for agricultural productivity.[5] We discard any Sentinel-2 scene with more than 50 percent cloud cover. Within the retained scenes, we mask individual pixels flagged as cloud, cloud shadow, or cirrus –

[4]The Modified Normalized Difference Water Index is defined as $\text{MNDWI} = \frac{\rho_{\text{Green}} - \rho_{\text{SWIR}}}{\rho_{\text{Green}} + \rho_{\text{SWIR}}}$, where ρ_{Green} and ρ_{SWIR} are reflectance in the green and shortwave-infrared bands.

[5]The Normalized Difference Vegetation Index is defined as $\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{Red}}}{\rho_{\text{NIR}} + \rho_{\text{Red}}}$, where ρ_{NIR} and ρ_{Red} are surface reflectance in the near-infrared and red bands, respectively.

thin high-altitude cloud – using the imagery's built-in scene-classification band. We then combine the surviving clear-sky pixels within each season by taking the median value at each location, which suppresses any residual haze, and we then average the clear pixels falling inside each field boundary to obtain one MNDWI and one NDVI value per field during the Rabi (i.e., dry) season of each year, aggregating readings from 1 November to 28 February each year. We chose the dry season to gain the cleanest differentiation between plots that are flooded with saltwater and those that are not flooded, but may have residual water from other uses. Plot-seasons based on fewer than three clear pixels are flagged as unreliable and dropped. Finally, a plot is classified as a shrimp pond in a given season when its MNDWI exceeds a season-specific cutoff chosen as follows. We classify each plot as a pond or agriculture by applying Otsu's method to the distribution of MNDWI values within each season and year, which selects the threshold that best separates the two modes of the distribution rather than imposing a fixed cutoff of zero.[6]

3.4 Satellite plot detection and classification quality

To conduct meaningful analysis using the satellite-derived panel, we need two conditions to hold: (i) the plot recognition algorithm is able to detect enough plots at a density that permits them to border one another frequently enough that we have sufficient variation in whether or not an agricultural plot is bordered by an aquaculture plot for an event-study approach; and (ii) the land use classification algorithm is not merely noise, i.e., it contains a sufficiently large signal about the actual use of that plot. Among the 51 unions that make up the satellite sample, areas designated as plots cover 36.8% of total space (see Table B1). Estimates place the share of land under cultivation at approximately 60%, implying that our delineated plots cover approximately 60% of cultivated land (Bari et al., 2024).[7]

Accurately classifying plots as agriculture or aquaculture is difficult in this setting because the water we are attempting to detect is shallow and murky, so errors may arise from either under- or over-classification of wetness. Appendix figure B4 shows the distribution of MNDWI readings for our sample by year. The vertical lines show the water threshold chosen each year by Otsu's method, which selects the

[6]Otsu's method chooses the threshold t^* that minimizes the intensity variance within the two groups it creates (equivalently, maximizes the between-group variance): $t^* = \arg\max_t \ \omega_0(t)\,\omega_1(t)\left[\mu_0(t) - \mu_1(t)\right]^2$, where $\omega_0(t), \omega_1(t)$ are the proportions of observations below and above t and $\mu_0(t), \mu_1(t)$ are their respective means.

[7]This percentage includes both the city of Khulna which has very little agricultural land as well as mangrove forests in the Khulna Division, so it is best interpreted as a lower bound.

cutoff that maximizes the between-class variance of the index. Although the index is "normalized", the selected threshold is above zero in every year, and many plots have intermediate values. Thus, although hereafter we refer to plots above the threshold as *ponds* or as being in *aquaculture*, the classification is more generally interpreted as placing a plot above a data-driven threshold of pond likelihood. As an alternative, we also classify plots using a single manual threshold, MNDWI > 0.241, held fixed across all years. We choose this value to minimize the balanced error rate—the average of the false-positive and false-negative rates—against land use reported in our household survey for plots we can match to the satellite data. Because it lies above the Otsu threshold in every year, the manual threshold applies a stricter definition of a pond. Our event-study results are robust to this reclassification.

4 A simple model of land use with an externality

This section sets out a model of land allocation between rice and saline aquaculture in which a plot's returns depend on whether its neighbors are in aquaculture. It separates selection on comparative advantage from the externality, and shows how the externality propagates across a landscape.

4.1 Environment

There are plots which are each operated by an individual farmer. Plots, indexed by i, are contiguous, and each is either in rice ($d_i = 0$) or in saline aquaculture ($d_i = 1$). Plot i has an unobserved productivity in each use, a_i^R and a_i^S which are independently drawn from Fréchet distributions. These capture everything that makes a plot more or less profitable in rice or shrimp, respectively beyond its observed characteristics: soil, drainage, access to fresh or saline water, and the skill and capital of whoever farms it.

A plot is exposed if it borders aquaculture:

$$b_i = \{\text{plot } i \text{ shares a boundary with at least one aquaculture plot}\}. \tag{1}$$

4.2 Payoffs and land use

Profit for each plot is productivity times observable shifters:

$$\pi_i^S = a_i^S \cdot \underbrace{e^{w_i'\gamma^S}}_{\text{characteristics}} \cdot \underbrace{e^{-c}}_{\text{conversion cost}}, \tag{2}$$

$$\pi_i^R = a_i^R \cdot \underbrace{e^{w_i'\gamma^R}}_{\text{characteristics}} \cdot \underbrace{e^{-\delta b_i}}_{\text{net damage}}, \tag{3}$$

where w_i contains plot size and distance to the river. Each plot is allocated to the use that pays more:

$$d_i = 1 \iff \underbrace{\log a_i^S - \log a_i^R}_{\Delta_i} > c - \delta b_i - w_i'\gamma, \qquad \gamma \equiv \gamma^S - \gamma^R. \tag{4}$$

Δ_i is the plot's comparative advantage in aquaculture. An aquaculture neighbor lowers the threshold by δ without changing Δ_i.

It is important to note that δ is a net damage effect: the loss of rice profit from an aquaculture neighbor plus any gain in shrimp profit from one. Both make aquaculture relatively more attractive, and land use responds only to their sum; writing δ on rice profit is a normalization. The multiplicative form makes the effect proportional, so a given exposure changes returns by the same fraction on every plot. The conversion cost c is common across plots. It cannot be separated from the intercept of $w_i'\gamma$ and is absorbed into it; plot-level variation in conversion costs is absorbed into Δ_i.

4.3 Types and allocation probabilities

Productivities are independent Fréchet draws with common shape θ, and unit scale since w_i enters explicitly:

$$\left(a_i^k \leq a\right) = \exp\left(-a^{-\theta}\right), \qquad k \in \{R, S\}. \tag{5}$$

Then Δ_i is logistic with scale $1/\theta$, and with Λ the logistic function,

$$P_i \equiv (d_i = 1 \mid b_i, w_i) = \Lambda\big(\theta(w_i'\gamma - c + \delta b_i)\big). \tag{6}$$

θ indexes the dispersion of comparative advantage. When it is low, plots differ widely and land use responds little to a change in returns; when it is high, plots are close substitutes and the same change reallocates more land.

As is standard with Fréchet types, realised profit $\pi_i = \max\{\pi_i^S, \pi_i^R\}$ has the same

distribution whichever use is chosen,

$$(\pi_i \leq x \mid d_i = k) = \exp\Big[- \left(e^{\theta(w_i'\gamma^S - c)} + e^{\theta(w_i'\gamma^R - \delta b_i)}\right) x^{-\theta} \Big], \qquad k \in \{R, S\}, \qquad (7)$$

so the dispersion of realised returns identifies θ.

Independence across plots is a substantive assumption. It rules out, for example, a single operator farming several adjoining plots, whose joint conversion the model would read as a cascade.

4.4 Interpretation

It is important to note that the model does not require the externality to be salinity. δ captures any channel through which an aquaculture neighbor changes a plot's relative returns, whether negative for rice, such as salinity, waterlogging or disease, or positive for shrimp, such as shared embankments and canals, information, or access to buyers and inputs. Changes in land use derived from the satellite data identify only the net effect. While this net externality is sufficient to simulate, the split between these channels matters for over- or underadoption and thus the correct policy response. We return to these concerns in Section 7.

The model is static. An equilibrium describes where land use settles given current returns, not the path or speed of adjustment, and plots respond to their neighbors without anticipating future conversions. We interpret it as a medium-run framework, net of climate change: productivity in both uses is held at current levels, so the model does not capture a secular decline in rice returns from rising salinity.

5 Stylized Facts

This section presents brief demographic information about the farmers and plots represented in the household survey. Then it introduces three stylized facts about the shrimp farming industry in Bangladesh.

Key features of the households surveyed are shown in Panel A of Table 1. Respondents are selected from among marginal plot operators (i.e., who are cultivating rice near other shrimp plots or cultivating shrimp near other rice plots).[8]. Both groups are of similar age, sex, and education. They are very concerned about climate change. On the 1–3 scale for climate change concern, almost all respondents in both groups considered climate change a "1 = very serious threat". Respondents engaged in

[8]For greater detail about sample selection, see Section 3

Table 1. Farmer and Plot Demographics

Panel A: Households			
	Shrimp farming	Non shrimp farming	p-value
Age	45.7	47.0	0.151
	(0.6)	(0.7)	
Male	0.970	0.962	0.591
	(0.009)	(0.010)	
Individual income (1,000 taka)	282.5	217.3	0.000***
	(12.5)	(10.2)	
Completed secondary school	0.297	0.258	0.238
	(0.023)	(0.024)	
Muslim	0.670	0.568	0.004***
	(0.024)	(0.027)	
Number of non-homestead plots	2.46	2.02	0.000***
	(0.05)	(0.06)	
Climate change concern (1–3)	1.24	1.21	0.341
	(0.02)	(0.02)	
Panel B: Plots			
	Shrimp pond	Agriculture	p-value
Plot size (decimals)	131.5	69.6	0.000***
	(7.5)	(3.4)	
How acquired			
Bought	0.071	0.091	0.182
	(0.012)	(0.010)	
Rented	0.478	0.384	0.001***
	(0.025)	(0.018)	
Inherited	0.349	0.444	0.000***
	(0.024)	(0.019)	
Year acquired	2012.6	2012.1	0.459
	(0.6)	(0.5)	
Water source			
Rainfed	0.482	0.746	0.000***
	(0.026)	(0.017)	
River	0.367	0.082	0.000***
	(0.026)	(0.011)	
Canal	0.296	0.029	0.000***
	(0.022)	(0.006)	
Pond	0.002	0.023	0.000***
	(0.002)	(0.005)	
Groundwater	0.379	0.716	0.000***
	(0.025)	(0.020)	
Plot count	496	1138	
Household count	394	345	

Notes: Means are reported with standard errors in parentheses. Panel A: robust standard errors; a household is classified as shrimp farming if it has at least one shrimp pond. Climate change concern is coded 1 = very serious threat, 2 = somewhat serious threat, 3 = not a threat at all. Panel B: cultivable/arable plots and shrimp ponds only; standard errors clustered by household. The last column reports the p-value of the difference in means between columns 1 and 2. One decimal equals 1/100 acre. Acquisition categories are not exhaustive (user-right, gifted, government land, and other are omitted), so rows do not sum to one. Water source categories are not mutually exclusive. The household count in Panel B is the number of households with at least one plot of that type. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

shrimp farming report 30% higher individual income (282.5 thousand taka annually for shrimp farming households vs 217.3 for non-shrimp farming households). This gap is also reflected in the number of plots operated by each group: shrimp farming households operate 2.46 non-homestead plots relative to non-shrimp farming households' 2.02. Shrimp farming households are 10 percentage points more likely to be Muslim. Panel B shows plot differences across plots that operate as shrimp ponds at any time of the year and those that do not. Shrimp ponds are much larger, averaging 1.32 acres relative to 0.70 acres for agricultural plots. Land allocated to either use is rarely purchased outright (7.1% for shrimp ponds vs 9.1% for agriculture). The two dominant forms of plot acquisition are renting (47.8% for shrimp and 38.4% for agriculture) as well as inheritance (35% for shrimp and 44.4% for agriculture). These plots have been operated by their current owners for, on average, 15 years at the time of the survey in 2025. There is no dominant water source for shrimp ponds, which use a combination of rainfall (48.2%), rivers (36.7%), canals (29.6%) and groundwater (37.9%). Agricultural plots use rainfall (74.6%), rivers (8.2%), canals (2.9%) and groundwater (71.6%).

Fact 1: The ground-truth salinity gradient exists

Figure 3 reports average soil salinity (green bars, dS/m) alongside a reading taken directly from the saline aquaculture pond (blue bar), pooled across all sample plots and measured in May–June 2026. The four bars are arranged by distance from the saline source. The leftmost bar is the reading inside the saltwater aquaculture ("wet") plot itself, which averages 10.3 dS/m. The two central bars are boundary readings taken on a plot that simultaneously borders a saline aquaculture plot and an agricultural plot: the wet/dry boundary (4.3 dS/m) and the dry/dry boundary (3.9 dS/m). The rightmost bar is taken from an agricultural ("dry") plot that borders only other agricultural land (3.4 dS/m). Two features stand out. First, there is a steep gradient between the aquaculture reading and the agricultural readings—salinity more than halves crossing from the pond to the nearest agricultural boundary. Second, among the agricultural readings there is a smaller but consistent declining gradient as one moves further from the saline plot, from 4.3 to 3.9 to 3.4 dS/m.

This pattern is suggestive evidence that local salinity on a farmer's land may be shaped by the land-use decisions of neighboring farmers, rather than by that farmer's own choices alone. An important caveat is that this is cross-sectional evidence, so the direction of causation is not identified. One competing explanation is selection on a pre-existing, static salinity gradient: if some areas were already saltier, those

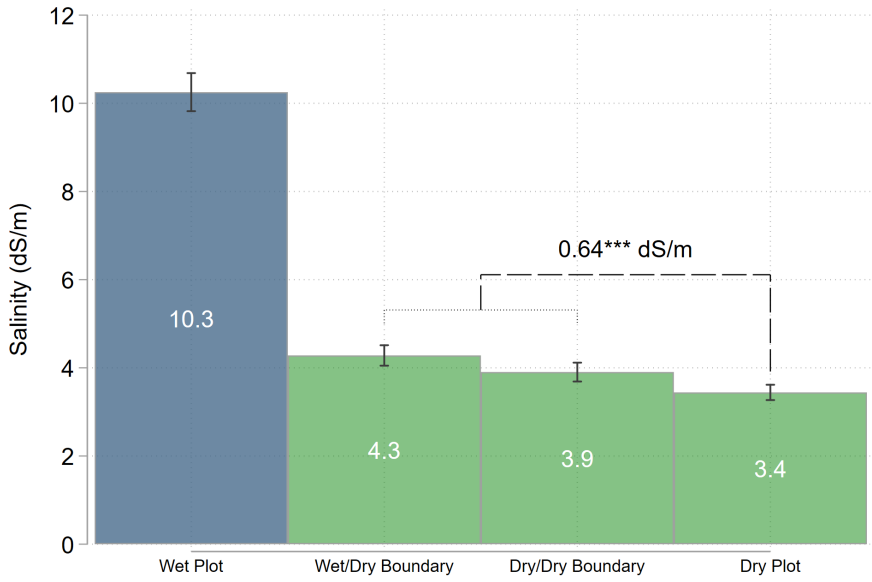


Figure 3. Salinity Readings Across Test Plots

plots may have been the ones converted to saline aquaculture, leaving the underlying spatial gradient—not the conversion—as the true driver of what we observe. That said, the agronomic literature makes lateral transmission of salt between adjacent plots physically plausible. Controlled monitoring and numerical modeling of onshore aquaculture farms document both vertical and lateral saltwater intrusion, showing that salt infiltration at ponds gradually invades neighboring farmland unless physically blocked by a guard wall, though evidence from Bangladesh is mixed (Ayers et al., 2017; Yu et al., 2024). Salt-balance studies of inland shrimp ponds similarly find that lateral seepage through pond walls and direct discharge of saline water into irrigation canals are the most significant salt-transfer pathways, raising salinity in adjacent canals and rice paddies above yield-damaging thresholds (Braaten and Flaherty, 2001). In this sense, salt can move across plot boundaries much as fertilizer or other soil inputs can spill over between neighboring fields. To distinguish this dynamic, spillover-based account from the static-gradient explanation, we supplement the cross-sectional comparison with the event studies below.

Fact 2: Saltier bordering plots have lower rice yields and profits relative to plots that do not border shrimp ponds, controlling for plot size

If proximity to a saline aquaculture pond raises the salinity of neighboring agricultural land, as Figure 3 suggests, then we should expect that elevated salinity to

depress the productivity of the affected plots. Figure 4 tests this directly, comparing annual rice yields—measured in kilograms per decimal (one decimal being 1/100 of an acre)—across agricultural plots according to whether they border a shrimp pond. Plots that do not border a pond yield approximately 23.5 kg/decimal, compared with 21.8 kg/decimal for plots sharing a boundary with a pond: a difference of −1.71 kg/decimal, or roughly seven percent, significant at the ten percent level. The salinity gradient documented in Figure 3 thus appears to carry through to agricultural outcomes—plots exposed to a neighboring pond are not only saltier but measurably less productive.

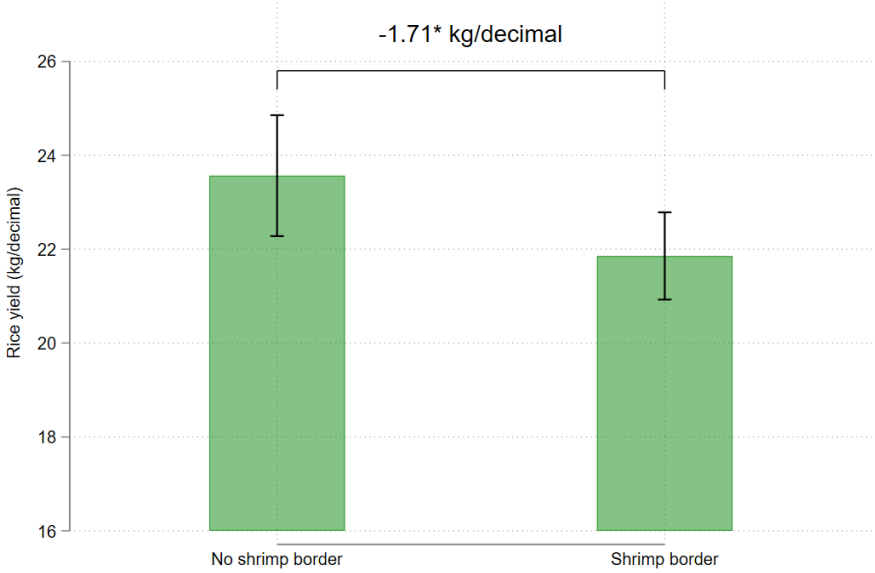


Figure 4. Rice Yields Across the Plot Distribution

As with the salinity readings, this comparison is cross-sectional and does not identify a causal effect. Border and non-border plots may differ systematically for reasons unrelated to the pond—in unobserved soil quality, location, or the characteristics of the farmers who operate them—and the same selection concern raised for the salinity gradient applies here: if plots that were already lower-yielding were the ones near which aquaculture was established, the yield gap would reflect pre-existing differences rather than the pond's influence. The figure should therefore be read as descriptive corroboration that the salinity spillover has real economic consequences, motivating the panel research design developed below, which uses the timing of neighbors' conversions to isolate the causal effect on land use and productivity.

Fact 3: Farmers understand these externalities

Figure 5 presents the distributions of two elicited outcomes—perceived water salinity (left panel) and subjective rice health (right panel)—under two scenarios. Both measures are reported on scales from 0 to 7. Salinity runs from 0 (no salt in the water) to 7 (the saltiest water the respondent has ever encountered). Rice health is an inverted scale: respondents are shown images corresponding to progressively poorer plant growth, so 0 denotes the healthiest rice and 7 the least healthy, with higher values reflecting the visible damage associated with greater salinity. The green bars show the status-quo distribution: for each plot, respondents report their current water access and their current rice condition. The red bars show the distribution under a hypothetical in which the plot bordering theirs is converted to saline aquaculture. In both panels, the distribution shifts markedly to the right when a neighboring plot converts—perceived salinity moves from a mass around 3–4 toward 4–6, and rice health shifts from the healthy end (concentrated at 2–3) toward the damaged end (piling up at 6–7). In other words, farmers expect that a neighbor's conversion will both raise the salinity of their own water and worsen the health of their rice.

This is important for two reasons. First, it shows that farmers understand the spillover process, at least directionally: their subjective expectations move in exactly the direction the agronomic evidence on lateral salt transmission would predict (Yu et al., 2024; Braaten and Flaherty, 2001), and they correctly link the intermediate channel (rising salinity) to the downstream outcome they care about (falling rice yields). Their beliefs are therefore not naive—they encode a coherent causal model of how a neighbor's land use propagates onto their own plot. Second, and consequently, these perceived spillovers are likely to enter farmers' own decisions. If a farmer anticipates that a neighbor's conversion will salinize their water and depress their rice yield, they may adjust their own cropping choices—or their own decision to convert to shrimp aquaculture—in response to what they expect their neighbors to do. This sets up a strategic, potentially self-reinforcing dynamic in which conversion decisions are interdependent across neighboring plots, which motivates the analysis of conversion behavior that follows.

In addition to this qualitative exercise, we ask farmers a hypothetical question in which they are told to consider a plot that is the same size as one of their randomly chosen existing plots that is growing rice and does not border any shrimp farms. When asked whether yields from that plot will change (from a benchmark of 100 kg) in the year following a neighbor's conversion to shrimp farming, they report an expected decrease in yields of 48%. This shows that the qualitative results from

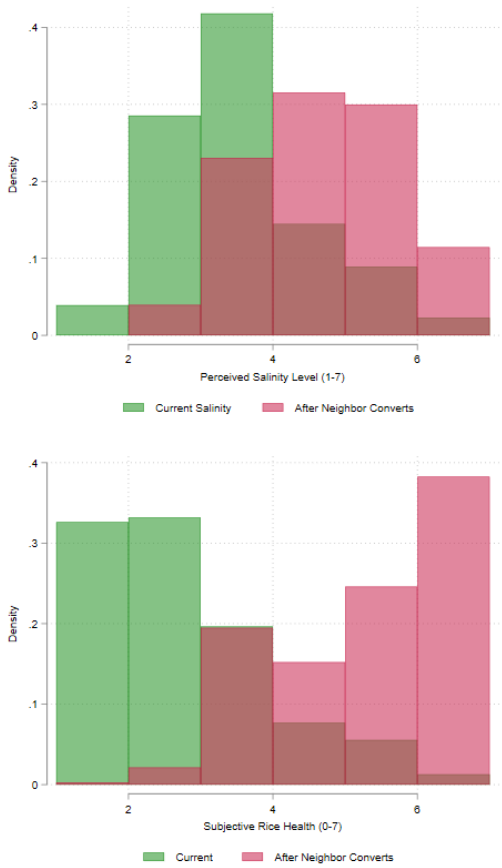


Figure 5. Salinity and Rice Outcomes After Neighbor Conversion

Figure 5 map onto a large quantitative decrease. We will use this quantitative gap to estimate the model of crop choice in Section 7.

6 Empirical Strategy & Event Study

We estimate whether a plot's conversion to shrimp aquaculture (hereafter a *pond*) induces its neighbors to convert. The central challenge is that adoption is spatially clustered for reasons unrelated to peer influence—soil salinity, water access, and proximity to processing infrastructure vary smoothly across space and jointly determine the profitability of aquaculture for neighboring plots. A naive comparison of conversion rates between plots with and without pond neighbors would conflate genuine contagion with these shared locational fundamentals. Our strategy exploits the *timing* of neighbors' conversions within a difference-in-differences design, comparing a plot's own adoption trajectory before and after a neighbor converts with that of plots

in the same small neighborhood whose neighbors have not converted.

The unit of observation is a delineated plot i observed in Rabi (dry) season t, for $t = 2017, \ldots, 2025$. Let $Y_{it} = 1$ if plot i is classified as a pond in period t and 0 otherwise. Each plot's treatment date E_i is the first period in which any plot sharing a boundary with i is classified as a pond. We estimate the event-study model:

$$Y_{it} = \alpha_i + \lambda_{h(i)t} + \sum_{k \neq -1} \beta_k \, D_{it}^k + X_i' \eta_t + \varepsilon_{it}, \tag{8}$$

where D_{it}^k indicates that plot i is k periods from the first period in which a neighbor is a pond, α_i are plot fixed effects, and $\lambda_{h(i)t}$ are hexagon-by-period fixed effects for the hexagon $h(i)$ containing plot i. The hexagons form a regular grid over the study area, sized so that each contains on average 20 plots of the estimation sample, which gives hexagons 1.03 km across; hexagons containing a single plot are dropped. X_i contains log distance to the nearest river and log plot area, each interacted with period so that their effects can vary over time. The period $k = -1$ is omitted as the reference, so each β_k measures the change in the probability that i is a pond at event time k relative to the period immediately preceding a neighbor's conversion. Extreme leads and lags are binned at ± 5 years, so that the endpoint coefficients are not identified off a single treatment cohort.

The plot fixed effects α_i absorb all time-invariant determinants of aquaculture suitability—soil type, elevation, baseline salinity, distance to water channels, and unobserved operator characteristics—so that identification comes entirely from within-plot changes over time. The hexagon-by-period fixed effects $\lambda_{h(i)t}$ absorb any shock common to a neighborhood of about 1 km in a given year, such as an advancing salinity front, local waterlogging, or shifts in local shrimp prices, as well as region-wide movements in prices and aggregate weather. Identification therefore comes only from differences in the timing of neighbors' conversions among plots in the same hexagon and year. The period interactions $X_i' \eta_t$ allow plots nearer rivers and larger plots to follow different trends. Because Y_{it} is binary, (8) is a linear probability model and the β_k are interpreted in percentage-point units.

A well-defined event study requires that treatment onset be observable within the panel. We therefore restrict the estimation sample in two ways. First, we exclude plots with no delineated neighbors, as neighbor exposure is undefined for them. Second, we drop plots that already border a pond in the first period of the panel. These plots are always-exposed: their treatment date is left-censored, and no clean pre-period exists against which to measure the effect of a neighbor's conversion. The retained sample

consists of plots that begin the panel with no pond neighbor, for which the first appearance of a pond neighbor is an observed transition rather than a pre-existing state. The estimand is accordingly local to the expansion frontier—plots at the margin of the aquaculture landscape—and does not describe the already-saturated core.

Because treatment timing is staggered and treatment effects may be heterogeneous across adoption cohorts, the two-way fixed-effects estimator of the β_k can be contaminated by comparisons in which already-treated plots serve as effective controls for later-treated plots; with dynamic effects, these comparisons enter the estimate with potentially negative weights (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020). We therefore report the interaction-weighted estimator of Sun and Abraham (2021), which estimates a separate coefficient for each adoption cohort and event time and aggregates them using each cohort's share of observations at that event time, so that all weights are non-negative. Never-treated plots—those whose neighbors never convert during the panel—form the control group. Because both land use and its unobserved determinants are spatially correlated, we compute standard errors following **?**, allowing for correlation across plots within 2 km of each other in the same period, with weights declining linearly in distance, and for arbitrary correlation within a plot across periods.

6.1 Identifying Assumption

The identifying assumption is parallel trends: absent a neighbor's conversion, treated plots would have followed the same own-adoption path as control plots. This is not directly testable, but its plausibility is assessed through the pre-treatment coefficients $\{\beta_k\}_{k<-1}$, which should be jointly indistinguishable from zero if treated and control plots were on common trajectories before treatment. The salient threat is a plot-specific, time-varying shock that drives both a neighbor's conversion and the plot's own—most plausibly a localized intrusion of saline water that renders a cluster of contiguous plots newly suitable for aquaculture in the same period. Such a shock would generate the appearance of contagion absent any causal peer effect. Time-invariant salinity differences are absorbed by α_i and region-wide salinity trends by λ_t; only *local, time-varying* salinity movements pose a threat.

6.2 Extensive Margin Effects

Figure 6 presents the event-study estimates for the probability that a plot is classified as a pond. Following a neighbor's conversion, a plot's own probability of being a

pond rises sharply and immediately, by 18.0 percentage points in the period following conversion. The effect persists across all subsequent periods, ranging between 0.14 and 0.22over the following five years, and shows no evidence of decay. Own-conversion thus appears to be a durable rather than transitory response: once a neighbor converts, the elevated likelihood of the plot itself becoming a pond is sustained rather than fading over time. The pre-treatment coefficients are flat and statistically indistinguishable from zero, consistent with the parallel-trends assumption.

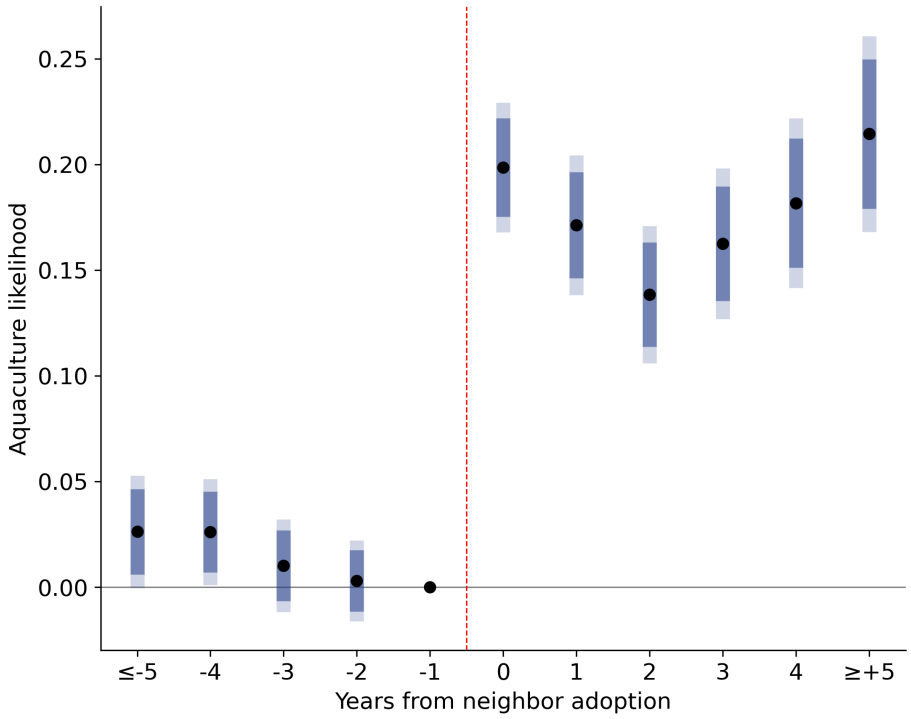


Figure 6. Aquaculture Event Study

Notes: The figure plots Sun and Abraham (2021) event-study estimates of own aquaculture status against years since a plot's first adjacent neighbor became a pond, for plots with no adjacent pond at baseline. The specification includes plot and hexagon × year fixed effects and river-distance × year and plot-size × year controls, and shaded bands show 95% (dark) and 99% (light) confidence intervals using Conley standard errors with a 2 km cutoff.

The effect emerges at event time zero—in the same period as the neighbor's conversion—rather than building gradually over subsequent years. This timing is informative about the mechanism. A gradual, increasing response would be consistent with social learning, in which a farmer observes a neighbor's outcomes over several seasons before updating and eventually converting. The immediate response we observe instead suggests that the second farmer acts concurrently with, or very shortly after, the neighbor's conversion. Such timing is consistent with an antici-

patory mechanism: if farmers believe that an adjacent pond will depress their own yields—for instance, through saline intrusion—the optimal response is to convert immediately rather than to wait and learn, since remaining in agriculture beside a pond becomes less viable as soon as the neighbor converts.

6.3 Intensive Margin Effects

Figure 7 presents the event-study estimates with NDVI—a satellite proxy for vegetative greenness, and hence crop productivity—as the outcome. This specification is estimated only on the sample of plots that had no pond neighbor at baseline and that never themselves convert to a pond at any point in the panel, so that NDVI always reflects standing agriculture rather than open water. Following a neighbor's conversion, greenness falls sharply and immediately: the effect is -0.019 at the time of conversion, indicating a contemporaneous decline in crop productivity. The effect then attenuates steadily over the following years, converging back toward zero by year five. The pre-treatment coefficients are flat and indistinguishable from zero, consistent with parallel trends. As with the extensive-margin result, the decline appears at event time zero rather than emerging gradually, indicating that the productivity cost of an adjacent pond—plausibly through saline intrusion—is felt immediately rather than accumulating slowly. The subsequent recovery toward baseline suggests that farmers who remain in rice may, over time, adopt mitigating practices that partially restore greenness to its pre-conversion level.

Two features of the sample warrant caution in interpreting these magnitudes. Because the estimation is restricted to plots that never convert, the sample is not representative of all plots initially exposed to a converting neighbor: precisely those plots that experience the largest productivity losses may be the ones most likely to abandon rice and convert to shrimp themselves, thereby exiting the sample. Conditioning on non-conversion therefore selects on an outcome that is itself affected by treatment, and it does so in a direction that biases the estimated harm toward zero—the worst-affected plots are absent. The coefficients reported here should thus be read as a conservative lower bound on the true productivity cost of a neighboring pond, capturing the effect among the subset of farmers who chose to remain in agriculture despite it.

6.4 Robustness

To examine how the effect of a neighboring conversion decays with distance, we re-estimate the event study separately by distance rings, assigning each plot to a

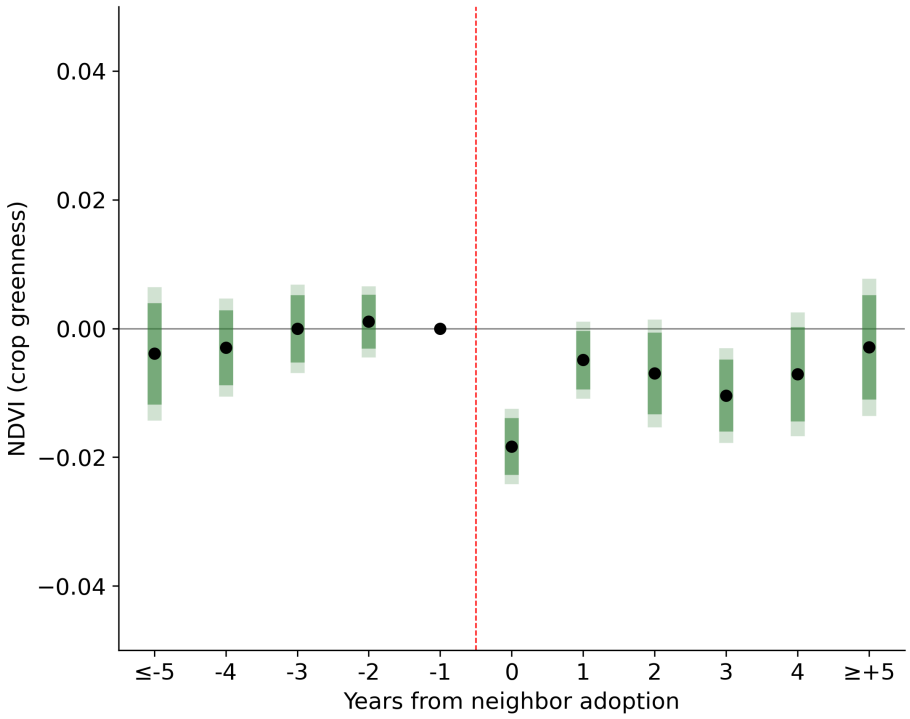


Figure 7. Vegetation Index Event Study

Notes: The figure plots Sun and Abraham (2021) event-study estimates of Rabi-season NDVI against years since a plot's first adjacent neighbor became a pond, for plots that are never classified as aquaculture and have no adjacent pond at baseline. The specification includes plot and hexagon × year fixed effects and river-distance × year and plot-size × year controls, and shaded bands show 95% (dark) and 99% (light) confidence intervals using Conley standard errors with a 2 km cutoff.

ring according to the distance between the plot and the first pond to appear within 500 meters. Appendix Figure B5 reports the results for four rings: plots whose first nearby pond appears within 20 meters (panel a), between 20 and 50 meters (panel b), between 50 and 200 meters (panel c), and between 200 and 500 meters (panel d). In each panel, the outcome is an indicator for whether the plot is itself a pond, and event time is measured relative to the year the first pond appears within 500 meters. The effect is concentrated among plots closest to the new pond. For plots within 20 meters, the probability of conversion rises by 18.0 percentage points on average over the post-period, with no evidence of differential trends before the neighboring conversion. Beyond immediate neighbors, we find no meaningful spillover. The average post-period effects are 0.24 percentage points for non-adjacent plots within 50 meters (SE 0.54), 0.23 percentage points between 50 and 200 meters (SE 0.25), and 0.62 percentage points between 200 and 500 meters (SE 0.22). All three estimates are

small, and their confidence intervals rule out effects larger than about one percentage point in either direction. The absence of an effect on nearby but non-adjacent plots is consistent with a spillover that operates only between immediate neighbors, as would be expected if saline water spreads primarily through seepage across shared boundaries.

6.5 Alternative Mechanisms

Our interpretation is that a neighbor's pond contaminates adjacent land with saline water, lowering the expected returns to rice and inducing conversion. We consider two alternative explanations for the conversion response.

6.5.1 Multiple Plots Converting at the Same Time Because of Shared Ownership

If adjacent plots belong to the same operator, an apparent neighbor effect could instead reflect a single decision applied across several of the operator's own fields. This is unlikely in our setting. Among surveyed shrimp farmers who cultivate more than one plot, the median distance between their own plots is 153 meters, compared with 19 meters between a plot and its neighbors. This is consistent with the fragmented land markets documented in southwestern Bangladesh (Haque, 2025), in which a single household's holdings are typically scattered rather than contiguous. Moreover, shared ownership would predict conversions concentrated in the same period as the neighbor's conversion, whereas the effects in our event study persist and remain large in the years after the neighboring pond appears.

6.5.2 Information Spillovers Between Plot Owners

Farmers may also learn about the profitability of shrimp from neighbors who have converted and follow them. Such learning likely plays some role: 16.8% of converters report that someone in their village told them shrimp farming was a good idea, compared with 1.6% who cite advice from an NGO or the government. However, two features of our results suggest that learning does not drive the effects we estimate. First, the effect appears immediately, in the year of the neighbor's conversion, before a neighbor's returns could be observed. Learning from a neighbor's experience would instead produce a response that builds over subsequent seasons. Second, information spillovers would not be expected to decline sharply with physical distance, since farmers report discussing shrimp farming more with landholders who do not border

their plots than with those who do ($p = 0.03$). By contrast, the effect we estimate is concentrated among plots within 20 meters of the new pond (Appendix Figure B5), as a physical spillover would predict.

7 Estimation and Counterfactuals

7.1 Estimation

The model in Section 4 is characterized by four sets of parameters, which we estimate by combining the household survey with the satellite panel. Table 2 summarizes the source of identification for each.

The dispersion parameter θ governs how heterogeneous plots are in their comparative advantage between rice and shrimp. Because productivity draws are Fréchet, θ is identified from the dispersion of log output value across surveyed plots. The slopes of the relative suitability index, γ, capture how plot size and distance to the river shift the relative returns to shrimp, and are estimated from a land-use logit on the survey sample. The survey cannot identify the level of γ, because marginal sampling selected plots on both sides of the rice–shrimp boundary and so excluded unexposed ponds. We therefore anchor the intercept to the satellite population, matching the share of plots without a pond neighbor that are in aquaculture (5%). This intercept absorbs the conversion cost c. Finally, we recover δ, the net effect of a pond neighbor on the relative return to shrimp, by matching the model's implied change in conversion probability to the event-study estimate in Section **??** ($\Delta P = 0.180$). We estimate $\theta = 1.95$ and $\delta = 1.27$.

An important feature of δ, as discussed above, is that it is a net effect. A pond neighbor may lower the return to rice through salinity damage, raise the return to shrimp through shared access to saline water and infrastructure, or both. Land-use choices depend only on the difference, so choice data alone cannot decompose δ into these components. As we show below, the decomposition does not matter for the positive predictions of the model, but it determines the sign of the optimal policy.

7.2 Exercise 1: Cascade Dynamics

We first use the estimated model to quantify how much of observed aquaculture is driven by the externality rather than by fundamentals. We simulate a stylized landscape of plots on a grid, draw productivities from the estimated distribution, and solve for the equilibrium allocation by iterating plots' best responses to their neighbors' land use, starting from the allocation implied by fundamentals alone. Because

Table 2. Model Parameters and Sources of Identification

Parameter	Estimated from	What it captures
θ	Survey: spread of log output	Dispersion of comparative advantage
γ (slopes)	Survey: land-use logit	How plot size and distance to the river shift relative suitability
γ (level)	Satellite: share of unexposed plots in aquaculture	Baseline adoption; absorbs the conversion cost c
δ	Satellite panel: event-study ΔP	Net effect of a pond neighbor

conversion is effectively irreversible and land use evolves from an initial state of rice cultivation, we select the smallest stable equilibrium.

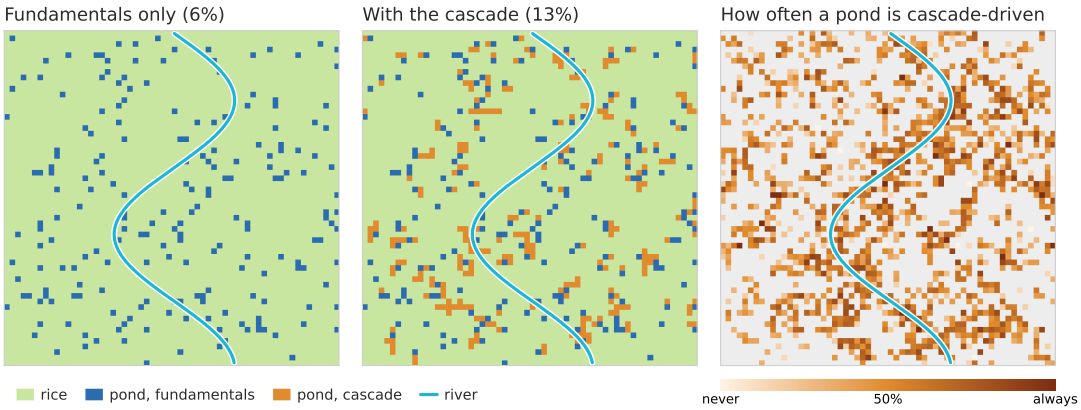


Figure 8. Shrimp Cascade on a Stylized Economy

Figure 8 simulates land use on a stylised 60×60 grid of plots, each with four neighbours and with plot size and distance to the river drawn from the observed distributions. Each plot draws its productivity in rice and shrimp from the estimated Fréchet distribution, with dispersion θ and covariate effects γ estimated from the survey and the level anchored so that the conversion rate among unexposed plots matches the satellite baseline. The left panel shows the allocation if every plot chose on fundamentals alone, ignoring its neighbours: 6% of plots become ponds. The middle panel adds the spillover: a converted neighbour shifts a plot's relative return to shrimp by δ, calibrated so the model reproduces the event-study effect of a neighbour's conversion, and choices are iterated to a fixed point. Aquaculture rises to 13%, and the orange plots are ponds that exist only because a neighbour converted first; they grow

outward from the fundamentals-driven ponds in blue. The right panel repeats the exercise over 40 productivity draws and shows, for each plot, how often it is cascade-driven when it is a pond (grey plots are ponds in fewer than 15% of draws). The exercise separates aquaculture that farmers would choose on their own land's merits from aquaculture that spreads through the externality, and shows that the cascade roughly doubles conversion, with about half of all ponds owing their existence to a neighbor.

7.3 Exercise 2: Optimal Policy

We next ask how a planner maximizing aggregate farm profit would intervene. The answer depends on a decomposition that land-use choices cannot reveal. The net effect of a pond neighbor combines two components,

$$\delta = \delta_R + \delta_S, \tag{9}$$

where δ_R is the damage that a neighboring pond imposes on rice and δ_S is the benefit it confers on a neighboring pond. Both components push an exposed plot toward aquaculture, so they are observationally equivalent in choice data, but they have opposite welfare implications. Damage to rice is a negative externality that calls for a tax on conversion, while a benefit to neighboring ponds is a positive externality that calls for a subsidy. The optimal policy is a tax when rice damage accounts for a large enough share of δ. In our calibration, this threshold corresponds to a rice yield loss of about 21% from a pond neighbor. At the other extreme, attributing all of δ to rice damage would imply a 72% yield loss.

We discipline the decomposition with two independent sources of evidence, which fall on opposite sides of the threshold. The satellite evidence in Section **??** implies a rice loss of about 9%, below the threshold. Relying on the satellite data alone would therefore attribute most of δ to benefits for neighboring ponds and call for a subsidy. Farmers' own assessments reverse this conclusion. Farmers in our survey expect a neighbor's conversion to reduce their rice yields by 48%, well above the threshold. If these beliefs are taken as expert knowledge of local conditions, the optimal policy is a tax, which reduces the aquaculture share to around 1% and raises aggregate farm profit by about 15%, capturing almost all of the gain available to an unconstrained planner.

We view the satellite estimate as a lower bound on the true damage for two reasons. First, NDVI measures greenness rather than yield, and salinity may reduce

grain yield more than vegetative growth. Second, the estimate is obtained from plots that do not convert, which are likely those least affected. The divergence between the two sources nonetheless shows that the welfare case for intervention, and even its sign, rests on the magnitude of damage to rice, which choice data alone cannot reveal.

8 Conclusion

This paper studies whether a private adaptation to climate change imposes costs on those who do not adopt it, and whether those costs accelerate adoption. In southwestern Bangladesh, where farmers can respond to rising salinity by converting rice paddies into saltwater shrimp ponds, we find evidence that they do. Rice plots bordering shrimp ponds are saltier and less productive than comparable plots, and farmers expect a neighbor's conversion to reduce their own rice yields substantially. Using a new plot-level panel of land use constructed from satellite imagery, we show that a neighbor's conversion lowers vegetation indices on adjacent agricultural plots and raises the probability that those plots convert by 32.6 percentage points. The effect appears immediately, is concentrated among plots within 20 meters of the new pond, and is difficult to reconcile with shared ownership or social learning. A model of land use with a boundary externality, estimated from the survey and the event study, implies that this response generates a substantial cascade: 73 percent of ponds in the simulated landscape exist only because a neighbor converted.

These results suggest that adaptation to climate change can be self-reinforcing in ways that are not captured by the usual focus on barriers to adoption. In many settings, frictions such as missing credit markets, risk, and limited information slow the take-up of new agricultural technologies. In ours, an externality on non-adopters works in the opposite direction, so that adoption spreads faster than comparative advantage alone would imply. Whether this is a problem depends on the size of the damage to rice. If the damage is as large as farmers believe, the market converts too much land to shrimp, and a tax on conversion would raise aggregate farm profit by about 15 percent. If the damage is closer to our satellite-based lower bound, the externality is dominated by benefits to neighboring ponds, and the case for intervention reverses. Because land-use choices alone cannot distinguish these cases, credible measurement of the damage that one farmer's adaptation imposes on another is central to the design of policy.

More broadly, the setting illustrates a feature of adaptation that matters beyond

coastal Bangladesh. In settings where adaptation to climate change will be decentralized, that is, carried out by households and firms responding to their own conditions rather than coordinated by the state, externalities can drive a wedge between private adaptation and the choice of the social planner. In these settings, evaluating adaptation by the gains to adopters alone will overstate its benefits, and the spread of a technology will be a poor guide to its social value. When rapid take-up can reflect the cost of remaining a non-user rather than the value of the technology itself, and policy that treats fast adoption as evidence of success risks encouraging more of it than is socially desirable.

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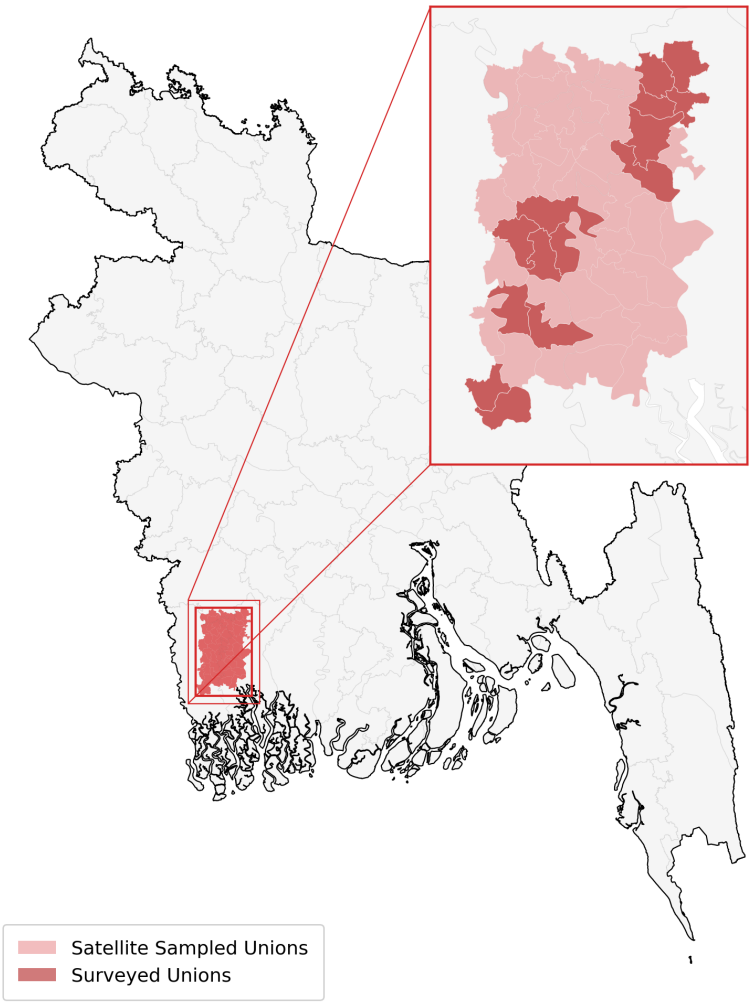
A Appendix 1 - Additional Tables

Appendix Table B1. Delineation Coverage by Union

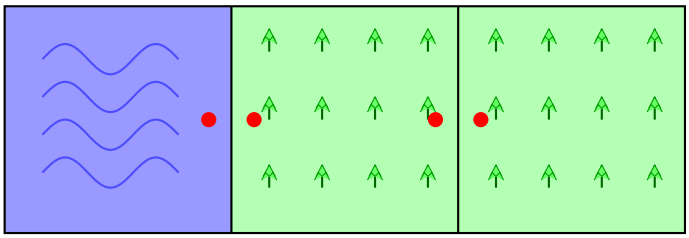
Union	Upazila	Area (km^2)	Area covered (km^2)	Share covered (%)	Plots
Anulia	Assasuni	28.82	10.54	36.6	1,202
Assasuni	Assasuni	34.86	13.64	39.1	1,528
Baradal	Assasuni	36.15	21.16	58.5	2,786
Budhhata	Assasuni	33.69	9.47	28.1	1,461
Durgapur	Assasuni	24.24	11.47	47.3	1,712
Kadakati	Assasuni	28.24	17.99	63.7	2,869
Khajra	Assasuni	37.29	18.80	50.4	1,934
Kulla	Assasuni	32.61	8.41	25.8	1,121
Sreeula	Assasuni	43.66	28.54	65.4	3,892
Atlia	Dumuria	42.01	13.52	32.2	1,506
Dumuria	Dumuria	18.68	1.34	7.2	238
Kharnia	Dumuria	21.21	2.21	10.4	509
Maguraghona	Dumuria	21.18	4.51	21.3	834
Magurkhali	Dumuria	41.66	22.75	54.6	3,359
Raghunathpur	Dumuria	34.88	12.45	35.7	1,852
Rudaghara	Dumuria	28.66	9.50	33.1	1,736
Sahas	Dumuria	25.07	7.90	31.5	1,406
Sobhana	Dumuria	45.45	7.48	16.5	918
Champaphul	Kaliganj	29.87	15.69	52.5	1,912
Krishnanagar	Kaliganj	26.22	10.10	38.5	1,800
Bidyanandakati	Keshabpur	32.76	5.69	17.4	1,051
Gaurighona	Keshabpur	25.82	4.83	18.7	1,070
Majidpur	Keshabpur	24.39	3.03	12.4	746
Mangalkot	Keshabpur	23.91	3.00	12.6	455
Sagardari	Keshabpur	28.79	3.71	12.9	1,068
Amadi	Koyra	41.87	19.95	47.6	2,744
Bagali	Koyra	48.32	23.05	47.7	2,157
Maheshwaripur	Koyra	52.03	14.35	27.6	771
Chandkhali	Paikgachha	45.67	23.52	51.5	3,657
Deluti	Paikgachha	46.09	24.72	53.6	1,433
Gadaipur	Paikgachha	20.39	8.95	43.9	860
Garuikhali	Paikgachha	42.86	28.10	65.6	1,352
Haridhali	Paikgachha	19.74	3.28	16.6	508
Kapilmuni	Paikgachha	38.01	16.92	44.5	2,233
Laskar	Paikgachha	42.16	24.92	59.1	2,516
Lata	Paikgachha	48.24	24.67	51.1	2,004
Paikgachha Paurashava	Paikgachha	4.81	0.73	15.2	56
Raruli	Paikgachha	25.36	7.74	30.5	922
Sholadana	Paikgachha	49.32	27.75	56.3	1,867
Dhulihar	Satkhira Sadar	37.95	16.68	43.9	2,078
Kashimari	Shyamnagar	34.40	16.38	47.6	1,378
Islamkati	Tala	22.18	3.38	15.2	819
Jalalpur	Tala	24.18	3.39	14.0	640
Khalilnagar	Tala	32.55	9.24	28.4	1,799
Khalishkhali	Tala	37.98	13.45	35.4	1,773
Khesra	Tala	48.02	10.83	22.6	1,066
Kumira	Tala	19.42	2.94	15.1	858
Magura	Tala	26.10	3.36	12.9	926
Sarulia	Tala	27.17	5.18	19.1	1,420
Tala	Tala	24.77	4.05	16.4	794
Tentulia	Tala	26.09	4.90	18.8	786
Total		1,655.82	610.15	36.8	76,382

Notes: The table reports the area of each union falling inside a delineated plot, and the share of the union covered. Union boundaries are from the Bangladesh Bureau of Statistics administrative boundaries. Plots are delineated from basemap imagery using Delineate Anything model described in Appendix C. Covered area is computed by clipping plots to union boundaries and merging overlapping polygons, so no land is counted twice. Plots are assigned to unions by the location of their centroid.

B Appendix 2 - Additional Figures

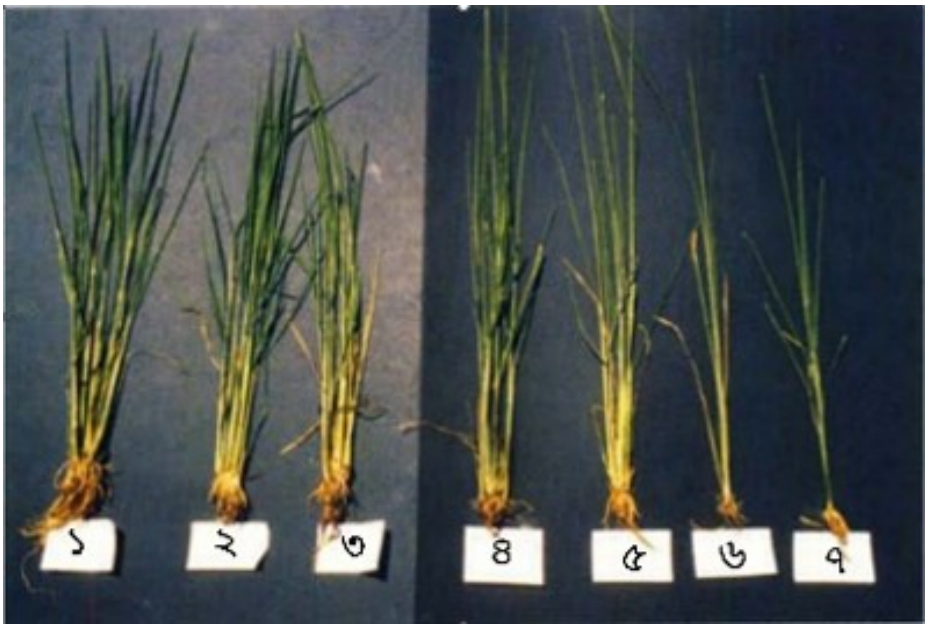


Appendix Figure B1. Sample Location

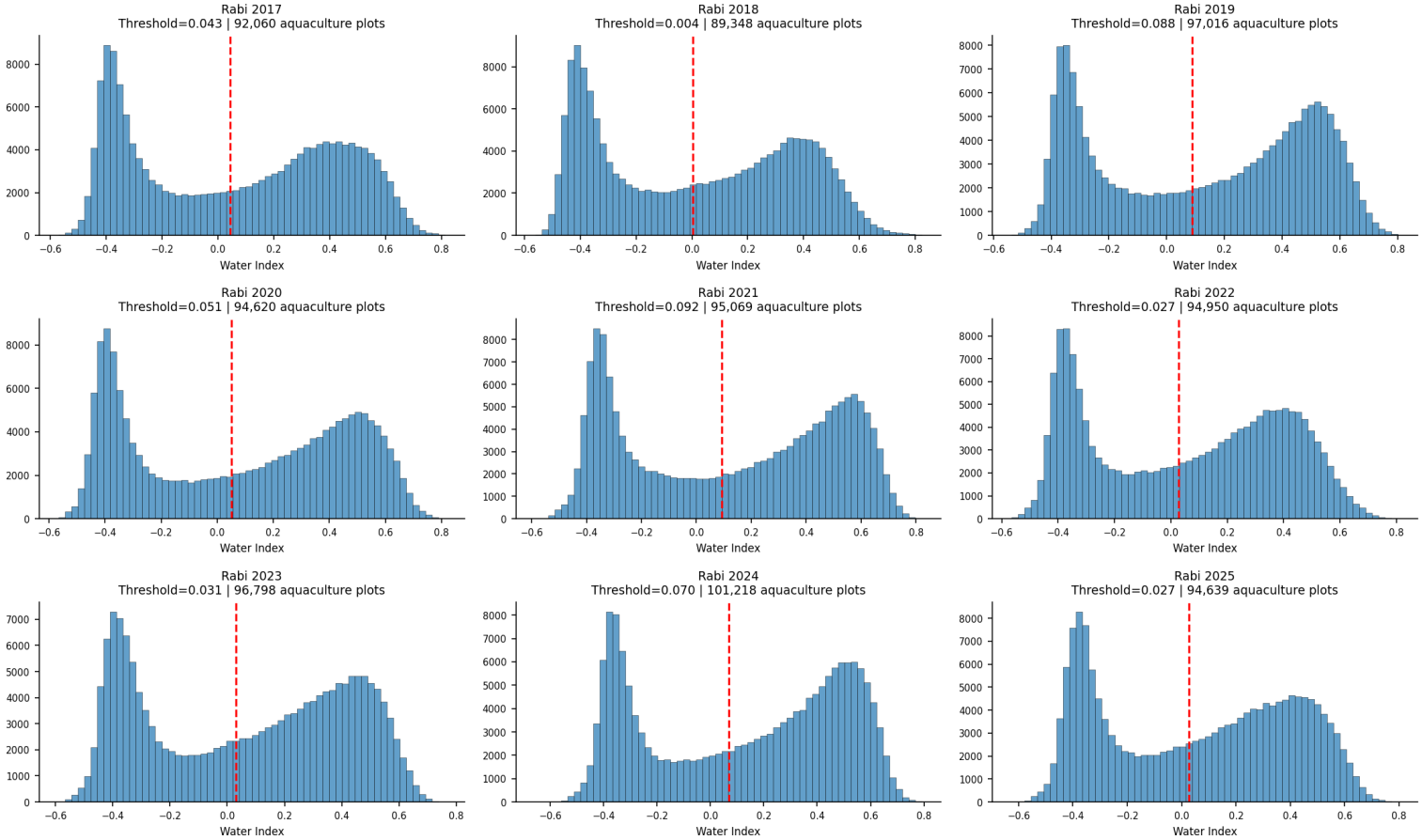


Appendix Figure B2. Salinity test diagram

Note: The diagram above shows an example of three plots, the leftmost of which is flooded with salt water for shrimp cultivation and the remaining two are used for agriculture. The red dots illustrate the approximate locations of the salinity readings.

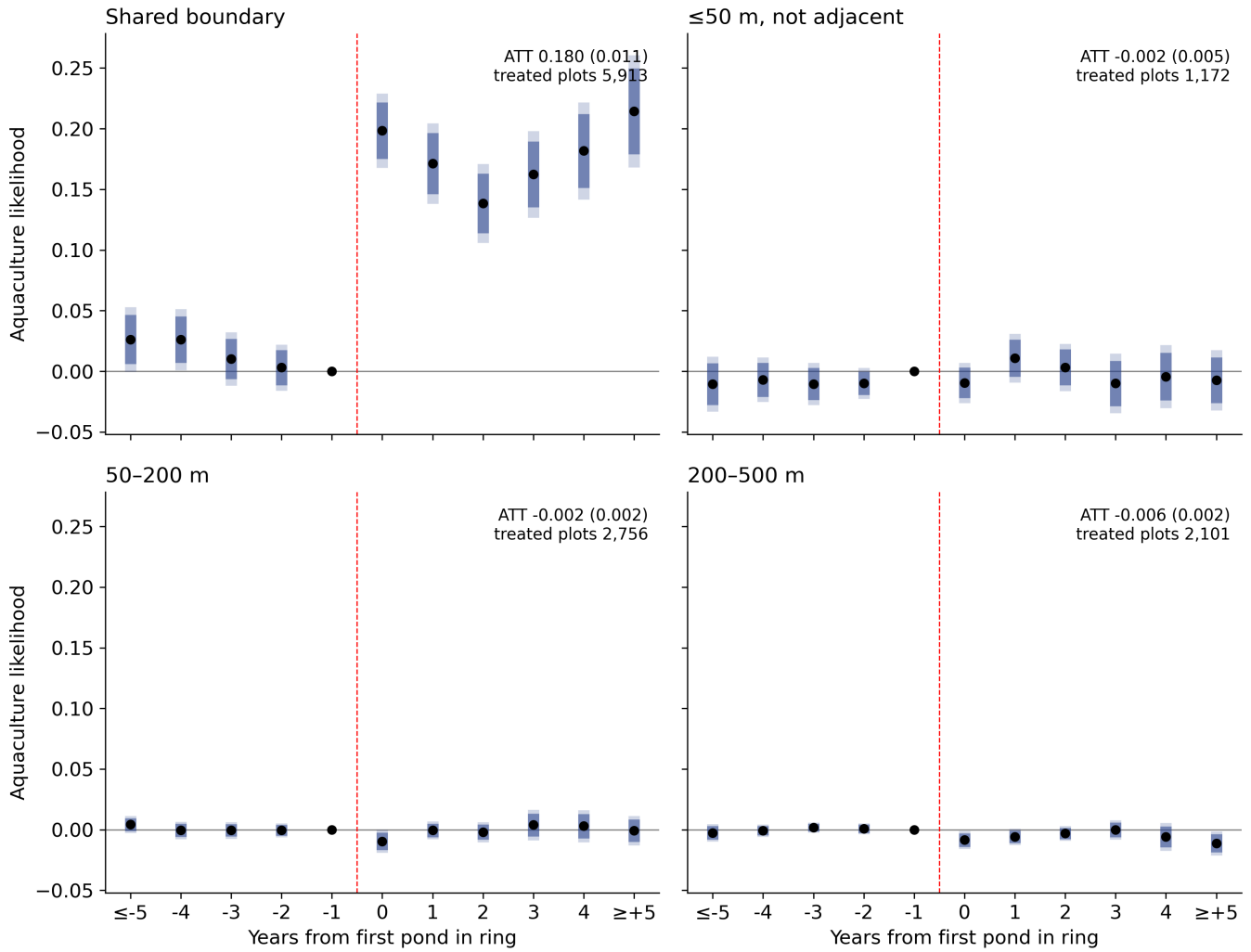


Appendix Figure B3. Rice Health Scale (from Grattan et al. (2002))



Appendix Figure B4. Modified Normalized Difference Water Indices Distribution

Notes: Each panel shows the distribution of the plot-level Modified Normalized Difference Water Index (MNDWI; Xu 2006) from Sentinel-2 imagery in the Rabi season (November–February), for plots with at least three valid pixels. The dashed line is that year's classification threshold, chosen by Otsu's method to maximize the between-class variance of the index; plots above it are classified as aquaculture, and the panel title reports how many.



Appendix Figure B5. Treatment Effects by Plot Rings

Notes: Each panel is a separate Sun and Abraham (2021) event study of own aquaculture status. Treatment is the first year a pond appears in that distance ring, measured edge to edge and excluding adjacent plots for the outer rings, and plot-years are censored once a closer ring is exposed. All specifications include plot and hexagon × year fixed effects and river-distance × year and plot-size × year controls. Shaded bands show 95% (dark) and 99% (light) confidence intervals using Conley standard errors with a 2 km cutoff.

C Appendix 3 - Delineate Anything Field-Boundary Algorithm

This appendix describes the procedure we use to convert raw satellite imagery into the individual plot polygons that serve as the units of analysis in the main text. We use *Delineate Anything*, an off-the-shelf, pre-trained machine-learning model for field-boundary delineation (Lavreniuk et al., 2025).

The goal is to take an aerial image of farmland and output a separate, closed polygon for each individual field. The engine inside DelAny is a neural network built on the YOLOv11 segmentation architecture, a widely used family of models designed to locate multiple objects in an image quickly and simultaneously (Lavreniuk et al., 2025). Rather than classifying loose pixels, the model is asked to identify each field as a distinct object and return that object's own filled outline (its "mask"). Because each field is detected as a whole entity, a small imperfection along one edge no longer merges it with its neighbor, which is the property we need for accurate plot-level measurement. DelAny was trained on the Field Boundary Instance Segmentation dataset (FBIS-22M), which contains roughly 673,000 satellite image patches and about 22.9 million hand-delineated field outlines. These training images span a wide range of spatial resolutions (from 0.25 to 10 metres per pixel), multiple satellite sensors, and several countries and seasons. Having learned from this deliberately heterogeneous set of examples, the model is "resolution-agnostic", that is it recognises the general visual signature of a field — a roughly convex patch of uniform texture bounded by abrupt changes in colour or tone — rather than memorizing the look of one specific sensor. This is what allows us to apply it to our imagery without retraining.

We apply the pre-trained model directly to our images without any further training or hand-labelling of Bangladeshi fields. The model requires only the three visible colour bands (red, green, and blue) as input, which our Planet basemap imagery provides. This is an advantage for transparency and replicability: our plot boundaries are the output of a fixed, publicly documented model applied to public imagery, with no discretionary labelling choices on our part.

Because sattelite imagery tiles overlap, the same field is sometimes detected more than once in adjacent tiles. A merging step reconciles these duplicates: when two detected outlines overlap by more than a set fraction of their area, they are treated as the same field and combined into one. This uses the standard notion of *intersection-over-union* — the ratio of the area two shapes share to the total area they jointly cover — as the measure of overlap, with a high threshold so that only near-identical detections are merged. After merging, the model's pixel masks are converted from filled raster blobs into clean vector polygons with explicit vertices, the format used by geographic information systems. Two size filters remove implausible detections: outlines smaller than a minimum ground area, and small spurious holes inside a field, are discarded.

Three caveats are worth noting for interpretation. First, the model can occasionally merge two truly distinct fields or split one field in two; because our analysis is at the plot level, such errors introduce measurement noise into plot boundaries. Second, fields at the outer edge of a quad can be clipped where two quads meet, which our cross-quad merging step mitigates but does not perfectly eliminate. Third, the boundaries are fixed from a single basemap and treated as static over the panel; genuine changes in field boundaries over time (for example, the subdivision or consolidation of plots) are not captured which may lead to measurement error.

Appendix Table B2. Delineate Anything Parameters

Parameter	Value	Default	Description
Detection			
Minimum confidence	0.005	0.005	Model confidence required to keep a detection
Minimum detection size	256 px	512 px	Smallest detected object retained, in pixels
Super-resolution	Auto	Auto	Optional upscaling before inference
Tiling and merging			
Tile step	0.5	0.5	Overlap between adjacent tiles (50%)
Merge IoU threshold	0.8	0.8	Overlap above which detections are merged
Tile composition IoU	0.8	0.8	Overlap above which detections across tiles are merged
Filtering			
Minimum field area	200 m^2	2,500 m^2	Smallest field retained
Minimum hole area	200 m^2	2,500 m^2	Smallest interior hole retained
Input and output			
Band order	[3, 2, 1]	[3, 2, 1]	Order in which image bands are passed to the model
Polygon simplification	Off	On	Simplification of output polygon vertices

Notes: Parameters used to run Delineate Anything (Lavreniuk et al., 2025) on Planet basemap quads. "Default" refers to the value in the model's sample configuration file. We lower the minimum detection size and minimum field area relative to the defaults because fields in southwestern Bangladesh are small; 200 m^2 corresponds to 0.02 hectares (about five decimals). IoU denotes intersection-over-union. Polygon simplification is disabled for computational stability. Computational settings (batch size of 16 and half-precision inference) do not affect the output and are omitted.

D Appendix 4 - Salinity Sensor Protocol

Section D.1 applies to the main plot, which was selected as a marginal shrimp or marginal rice plot and used to construct the sample as shown in Figure B2. Section D.2 applies to one randomly selected additional plot from each household's plot roster for additional salinity testing. Enumerators were provided with handheld electrical conductivity salinity meters with a range 0∼10.00ms/cm, resolution 0.01ms/cm and Accuracy ±2%F.S. Each meter was calibrated prior to the survey using the standard calibration solution.

D.1 Main Plots

After completing the interview with the respondent, please return to the main plot and follow the steps below for aquaculture/agriculture boundaries. For agriculture/agriculture boundaries, begin at step 3 and repeat steps 3 and 4 twice instead of beginning with step 1. Rinse the sensor with bottled water and dry with a clean cloth after each reading.

1. Measure the salinity of the water at the boundary of the aquaculture plot. Submerge the entire probe of the salinity sensor. Photograph the sensor display and record the reading on the tablet.
2. Take a second reading at the same boundary, photograph the display, and record the reading on the tablet.
3. Return to the border between the aquaculture and agricultural plots and walk five paces into the agricultural plot. Insert the sensor into the soil, photograph the display, and record the reading on the tablet. Wipe the sensor clean.
4. Move five paces to the left, parallel to the border, and repeat step 4, for a total of three soil readings.
5. Record an estimate of the width of the central agricultural plot.
6. Record whether it was raining at any time during the interview.

D.2 Other Plots from the Plot Roster

For each additional agricultural plot listed in the plot roster, the enumerator follows the steps below. Rinse the sensor with bottled water and dry with a clean cloth after each reading.

1. Walk five paces into the agricultural plot. Insert the sensor into the soil, photograph the display, and record the reading on the tablet.
2. Move five paces to the left and repeat step 1 for a total of two readings.